

CodHer'26

TEAM: SheCodes

Real Time Ambulance Routing & Traffic Clearance System

Saving Minutes. Saving Lives.



1. HARIPRIYA M



2. KAVIYA S



3. MERLIN M



IDEA

OVERVIEW



What We're Building

A system that helps ambulances reach hospitals faster by clearing traffic and alerting hospitals in advance.



Smart Signal Control

Auto-clears traffic signals along the ambulance route in real-time



Live GPS Tracking

Pinpoint ambulance location with optimal route calculation



Hospital Pre-Alert

ER teams notified minutes before ambulance arrival

PROBLEM STATEMENT



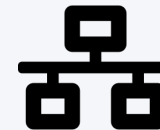
Traffic Gridlock

Ambulances get stuck in traffic, losing critical minutes during emergencies. No automated signal override exists in most Indian cities.



No Advance Warning to Hospitals

Hospitals aren't informed in advance, leading to unpreparedness. ER teams scramble last-minute, causing treatment delays.



Slow Manual Coordination

Existing systems are manual and fragmented. Ambulance, traffic, and hospital communication is siloed, not integrated.

Every minute saved in emergency response can mean the difference between life and death

PROPOSED SOLUTION

01



Real-Time Tracking

GPS-based live ambulance tracking with optimal route computation using traffic-aware algorithms.

02



Auto Signal Priority

IoT sensors + cameras trigger automatic green light cascade along the ambulance corridor.

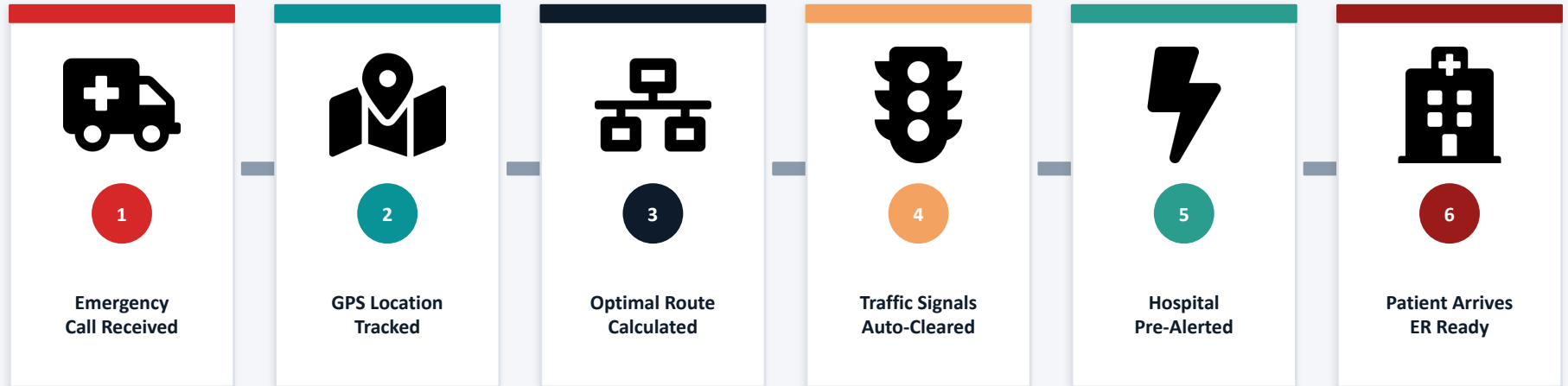
03



Hospital Pre-Alerts

System auto-notifies ER teams and reduces wait. Also flags traffic congestion in advance.

PRODUCT WORKFLOW



Integrated IoT + GPS + Cloud — Seamless end-to-end emergency coordination

CUSTOMER SEGMENT



Emergency Patients & Ambulance Staff

Direct beneficiaries of faster response



Hospitals & ER Teams

Pre-alerted for faster patient intake



Government & Smart City Bodies

Smart city integration partners



Traffic Control Departments

Automated signal management users

VALUE PROPOSITION



Faster Emergency Response

Reduce ambulance travel time by up to 40%



Reduced Mortality Rate

Save lives through timely critical care



Real-Time Coordination

Live sync between ambulance, traffic & hospital

REVENUE MODEL

₹80 Cr

TAM
(Tamil Nadu)

₹16 Cr

SAM
(Smart Cities)

₹30 L

SOM
(Chennai)

₹500

SaaS Fee
/pole/month

Unit Economics per Smart Pole

Component	Cost (INR)
Hardware (ESP32, LoRa, Radar, Solar)	₹2,800
Software & Cloud Allocation	₹500
Installation & Assembly	₹700
Total Cost per Unit	₹4,000

Revenue Streams



SaaS Subscriptions

₹500/month per smart pole



Government Contracts

Municipal & Smart City deployments



Hospital Onboarding

Premium hospital alert service tiers

COMPETITIVE ANALYSIS

Google Maps

✓ Strengths

- Wide coverage
- Real-time traffic data
- Route optimization

✗ Gaps

- No ambulance priority
- No signal control
- No hospital alerts

RapidSOS

✓ Strengths

- Emergency dispatch integration
- Location sharing

✗ Gaps

- No traffic signal override
- Not designed for India
- No hospital pre-alert

Our System

✓ Strengths

- End-to-end integration
- Auto signal override
- Hospital pre-alerts
- India-first design

✗ Gaps

- Early stage / MVP

We bridge the gap no existing solution fully addresses — a fully integrated, India-ready system.

SheCodes

Team Building the Future of Emergency Response

THANK
YOU

Real Time Ambulance Routing & Traffic Clearance System

CodHer'26 | Round 1 | Team SheCodes

Saving Minutes. Saving Lives.